

GuardShield AI — Catalog of Published Research Datasets

Department: CSE (AIML), Amity School of Engineering & Technology (ASET) | **Group:** 50 | **Guide:** Dr. Abhishek Kaushal

Every benchmark dataset used in GuardShield AI originates from top-tier peer-reviewed research papers published at flagship conferences (NeurIPS, ICLR, EMNLP, ACL, NAACL). Below is the consolidated reference matrix for research publication and faculty review.

Dataset	Focus Area	Conference & Year	Publishing Institution / Authors	Official DOI
JailbreakBench	Prompt Injection / Jailbreak	NeurIPS 2024	UPenn (P. Chao, A. Robey et al.)	10.52202/079017-1745
HackAPrompt	Adversarial Injections	EMNLP 2023	Univ. of Maryland & Scale AI	10.18653/v1/2023.emnlp-main.982
AdvGLUE	Adversarial Robustness	NeurIPS 2021	UIUC & Microsoft Research	10.52202/060246-0315
Do-Not-Answer	Extreme Model Risk	NeurIPS 2023	MBZUAI & Univ. of Melbourne	10.52202/075280-0442
AutoDAN	Stealthy Jailbreaks	ICLR 2024	UW-Madison & UC Davis	10.48550/arXiv.2310.04451
BIPIA	Indirect Injection	arXiv / MS 2023	Microsoft Research (J. Yi et al.)	10.48550/arXiv.2312.14197
BeaverTails	Safety & Benign Baseline	NeurIPS 2023	Peking University & ETH Zurich	10.52202/075280-1072
HaluEval	Hallucination Benchmark	EMNLP 2023	Renmin Univ. & Univ. de Montréal	10.18653/v1/2023.emnlp-main.397
TruthfulQA	Factuality & Falsehoods	ACL 2022	Univ. of Oxford & OpenAI	10.18653/v1/2022.acl-long.229
FEVER	Fact Extraction & NLI	NAACL 2018	Univ. of Cambridge & Amazon	10.18653/v1/N18-1074
FaithDial	Dialogue Hallucination	ACL 2022	McGill Univ., Mila & Univ. of Alberta	10.18653/v1/2022.trans-acl.1.72
KoLA	World Knowledge Grounding	ICLR 2024	Tsinghua University (J. Yu et al.)	10.48550/arXiv.2306.09296

Summary of Core Published Contributions:

- Pre-Scan Defense (Injections & Jailbreaks):** Uses standardized adversarial prompts from *JailbreakBench* (NeurIPS '24), *HackAPrompt* (EMNLP '23), and *AdvGLUE* (NeurIPS '21) augmented with Base64, ROT13, and Hinglish code-mixing.
- Post-Scan Defense (Hallucination Detection):** Employs *FEVER* (NAACL '18) for Cross-Encoder NLI factual grounding and *TruthfulQA* (ACL '22) & *HaluEval* (EMNLP '23) to benchmark token logit Shannon entropy spikes.
- Safe Baseline Calibration:** Ingests *BeaverTails* (NeurIPS '23) safe developer prompts to calibrate high precision (86.96%) and eliminate false blockage of legitimate user queries.